

Problem

Efficient Transformer attention mechanisms have proliferated, but **no single one is best across tasks** — so practitioners lack a cheap way to pick the right attention for a long-range task.

Attention performance is highly task-dependent when models train from scratch, without pretraining — the choice stays unclear.

Bigbird

Linear

Linformer

Local

Longformer

Performer

Reformer

Sparse

Synthesizer

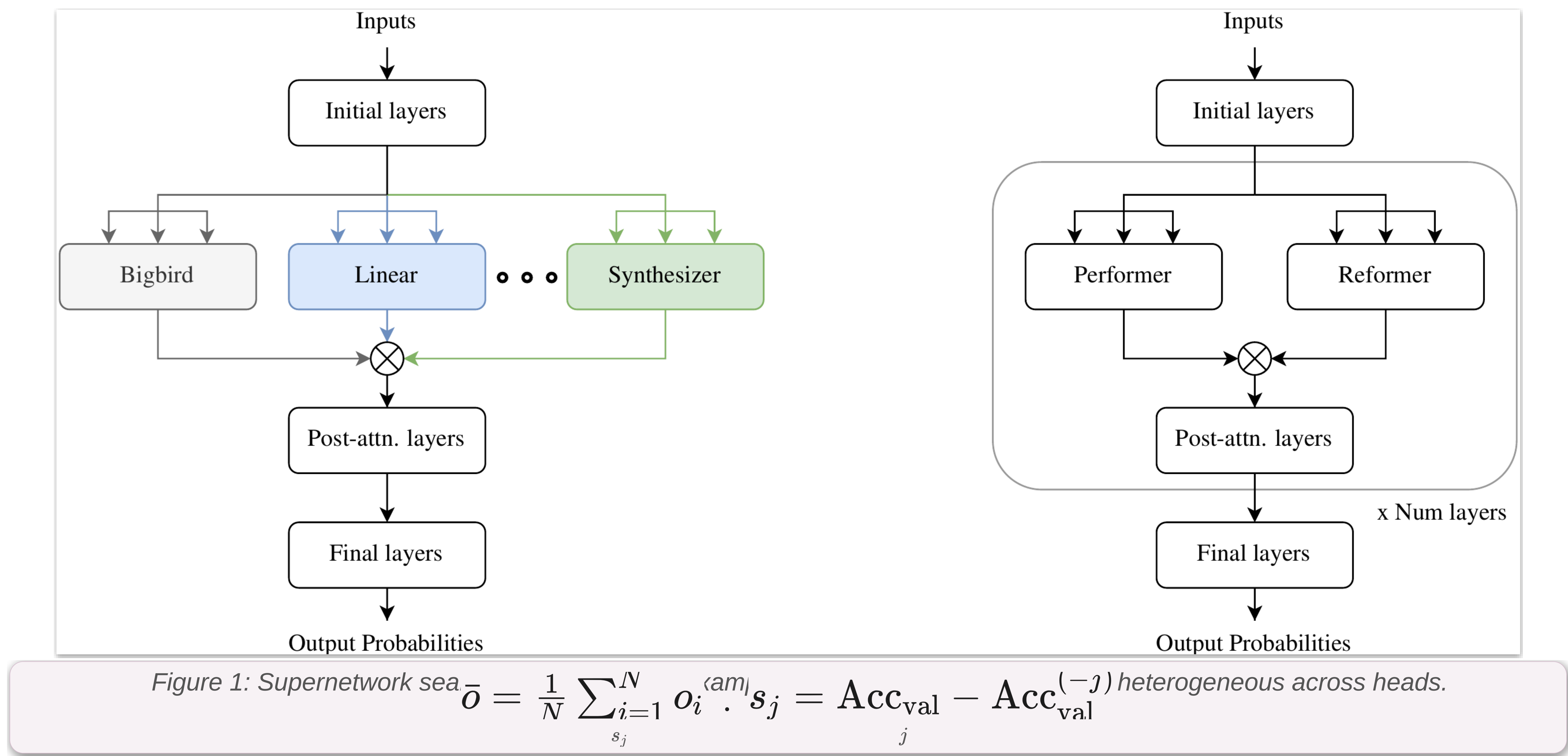
Motivation

- Attention heads are widely believed to learn **different relational biases** — so a *mixture* of attention types might beat any single one.
- Brute-force training of every attention (and every combination) is prohibitively expensive, motivating a cheap **search-based** selection.
- Much as convolutional kernels learn different features, specialized heads could compound *complementary* biases into a richer model.

Is the best attention for a task actually a mixture? DARTFormer tests this directly.

Method

- Build a single-layer **supernetwork** holding one candidate block per efficient attention type, trained to convergence.
- **Fixed- α DARTS**: average the block outputs (no softmax edges, no bi-level optimization), then score each by its **masked validation accuracy drop**.



Dataset / Benchmark

- Three **Long Range Arena** tasks: IMDb byte-level classification, ListOps 10-way, and document matching.
- Search space of **9 efficient attentions** (Bigbird, Performer, Reformer, Linformer, ...); each model trained 3× and averaged.

IMDb 1k

ListOps 2k

Doc-match 4k

Key Results

Task	Best Homog.	Prune	1-shot
Text classif.	64.5	63.9	64.4
Doc. matching	71.1	67.0	64.7
64.5 text · Performer	37.8 ListOps · Reformer	71.1 doc · Synthesizer	

SO WHAT → Search reliably finds the best single attention, but mixing types beats only the average — never the best.

Ablation Study

- **NAS Prune** (worst-first pruning) gives **no consistent gain** over cheaper **NAS One-shot** (top-4, one pass); heavier weighting of strong attentions also failed.

SO WHAT → Once good attentions are found, cheap one-shot selection suffices.

Headline Numbers

11.85

Reformer drop-score · ListOps

9 attentions

3 LRA tasks

0 hetero wins

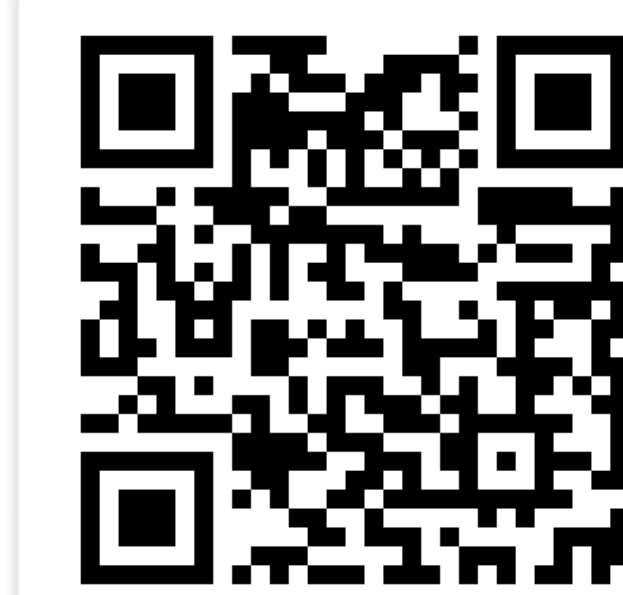
SO WHAT → Decisive on ListOps; heterogeneous never ahead.

Takeaway

TAKEAWAY
A cheap masked-validation-drop search reliably finds the best single attention — but combining types beats only the *average* homogeneous model, never the best. Prefer one well-suited attention over a mixture.

Questions? Get in touch

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Paper: arxiv.org/abs/2210.00641



Paper